

Dynamic Modeling and Simulation on a Hybrid Power System for Dual-motor-drive Electric Tracked bulldozer

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Abstract. Hybrid power systems provide high-performances and high-efficiency power for electric bulldozer. This paper firstly constructs the mathematical model of driving motors system based on the design of dual-motor-driving propulsion system for bulldozer. Secondly, by using the setting values from a bulldozer, simulation work is implemented, and parameters of the hybrid power system are matched, and finally, based on MATLAB/Simulink, dynamic models for the dual-motor-driving hybrid power system is established, and the simulation results are discussed. Simulation results demonstrate the validity of the hybrid power system model and the matching of the parameters of the hybrid power system is reasonable.

Introduction

With the social development, the number of bulldozer is increasing significantly. At the same time, the negative effects of the ever-increasing number of bulldozer have adversely impacted both energy sustainability and overall air quality. Even worse, the energy generated from oil yields the lowest energy efficiency, with an approximate rate of only 20%. Caterpillar in the United States produced a revolutionary D7E electric drive track-type bulldozer with electric-drive system in March, 2008. Compared with internal-combustion driving track-type bulldozers, D7E can yield greater fuel efficiency by nearly 10% to 30%, and at the same time enhance drive train efficiency and take lower life-cycle maintenance costs.

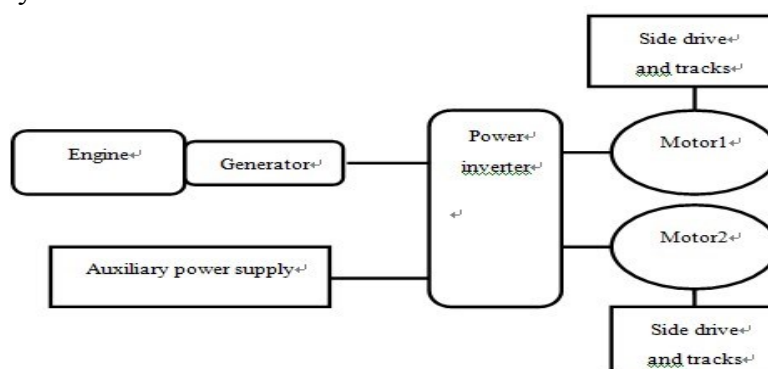


Figure 1. Structure of dual-motor-drive electric bulldozer

As showed in fig.1, the paths of electric power and mechanical power are tandem to each other in the series configurations, respectively.

Dual-motor-drive electric bulldozer uses the integrated controller to control the motors on the both sides of the bulldozer independently and transfers electric energy from the generator and auxiliary power supply into mechanical energy in order to drive the bulldozer.

Mathematical Models

Power, Torque, rated and maximum speed are important factors in parameter matching of driving motors system. These parameters are considered specially during driving motors' mathematical modeling progress [1].

Maximum Speed $n_{\max}$. The selection of maximum speed mainly based on the highest velocity of the bulldozer, actual motor design capabilities and the size requirements. Maximum speed and side gear ratio should meet the following relationship.

$$n_{\max} = \frac{u_{\max} i}{0.377r} \quad (1)$$

where r is radius (m); i is side gear ratio; $u_{\max}$ is maximum velocity of the bulldozer.

Rated Speed n_e . Rated speed should be the long-term continuous operation speed point. Therefore, the rated speed should be matched according to conventional operation speed of the bulldozer.

$$n_e = \frac{u_e i}{0.377r} \quad (2)$$

where u_e is bulldozer normal operation velocity.

Motor's Torque T . As the dual-motor-drive power system, T here discussed is single motor's torque. Motor's torque should meet power need during the whole working progress.

In the whole working process, the bulldozer's speed is less than 6km/h and the acceleration is small, so the air resistance and the inertial resistance are ignored when the mathematical models is established.

The single motor's torque is given according to the follow states and relationships [2].

$$T = \frac{(fG + F_T)r}{2i\eta} \quad (3)$$

where f is the rolling resistance coefficient; η is the efficiency from the traction motor to the track; F_T is the operating resistance (N);

Operating resistance (F_T) can be given as

$$F_T = F_1 + F_2 + F_3 + F_4 \quad (4)$$

$$F_1 = 10^6 B_1 h_p k_b \quad (5)$$

$$F_2 = G_t \mu_1 \cos \alpha = \frac{V \gamma \mu_1 \cos \alpha}{k_s} \quad (6)$$

$$F_3 = 10^6 B_1 X \mu_2 k_y \quad (7)$$

$$F_4 = G_t \mu_2 \cos \delta^2 \cos \alpha \quad (8)$$

$$V = \frac{B_1 (H - h_p)^2 k_m}{2 \tan \alpha_0} \quad (9)$$

where F_1 is soil-cutting resistance (N); F_2 is pushing resistance of mound before blade (N); F_3 is frictional resistance between ground and blade (N); F_4 is horizontal component of frictional resistance while soil rises along blade (N); k_b is cutting resistance per unit area (Mpa); B_1 is dozer's

blade width (m); h_p is average cutting depth (m); G_t is the gravity of the mound before bulldozing plate; μ_1 is friction coefficient between soil and soil; μ_2 is friction coefficient between soil and blade; α is slope ($^\circ$); γ is gravity per cubic meter (N/m^3); V is volume of the mound before bulldozing plate; k_s is loose degree coefficient of soil; k_m is fullness degree coefficient of soil; H is

dozer blade height (m); α_0 is the natural slope angle of the soil ($^\circ$); K_y is cutting resistance per unit area after blade pressed into the soil (Mpa); X is length of the worn blade contacting with ground (m); and δ is cutting angle of the dozer blade ($^\circ$).

Motor's Power. Motor's torque, power and motor's speed should meet the following relationship.

$$T = 9549 \frac{P}{n} \quad (10)$$

Simulation. Based on the motor's mathematical model above, the motor's parameters are simulated for a bulldozer. The main simulation parameters of bulldozer and working conditions are set in table. 1.

TABLE I. SIMULATION PARAMETERS FOR DUAL-MOTOR-DRIVING BULLDOZER

η	G (kg)	H (m)	B_1 (m)	k_s	L (m)
0.86	28 000	1.58	3.54	1.06	3.05
h_p (m)	k_b (MPa)	μ_1	μ_2	f	X (m)
0.2	0.06	1.0	0.5	0.1	0.01

In order to change the bulldozer's structure as little as possible, original side gear ratio i is adopted in this mathematical model of the motor.

Based on the above motor's mathematical model, the driving motor's parameters for a dual-motor-drive electric bulldozer are simulated. Simulation result is shown in fig.2.

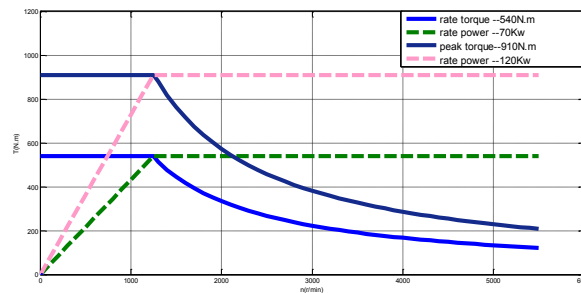


Figure 2. Single motor's characteristic curves

Dynamic Modeling for Hybrid Power System

Model of the Engine. The diesel engine universal characteristic curve is shown in fig.3. As shown in fig.3, the engine's lowest fuel consumption working points range from 1200rpm to 1500rpm. The engine's fuel injection quantity is adjusted by rack position [3]. The torque is obtained by the position of rack position and the speed of the engine. The model of the engine is shown in fig. 4.

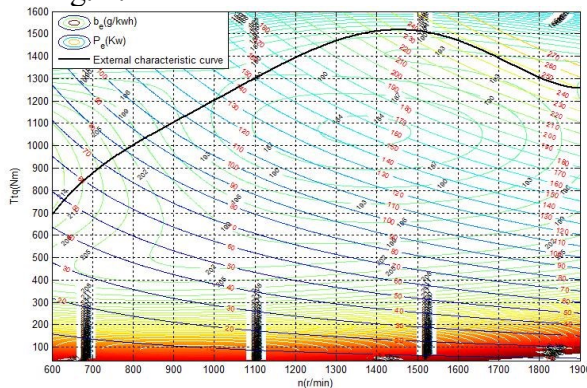


Figure3. The diesel engine universal characteristic curve

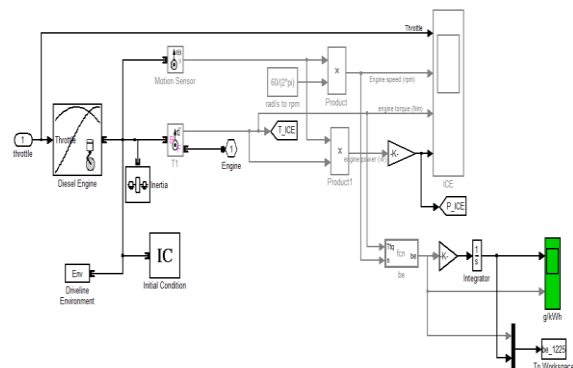


Figure4. Model of the Engine

Model of the Generator. The model of generator-rectifier is established by equivalent circuit method [4]. The generator can be regarded as a controllable voltage source in the electrics. DC bus voltage is related to DC bus current. The model is shown in Fig. 5. The inputs are engine torque and DC bus current; the output is the electromotive force.

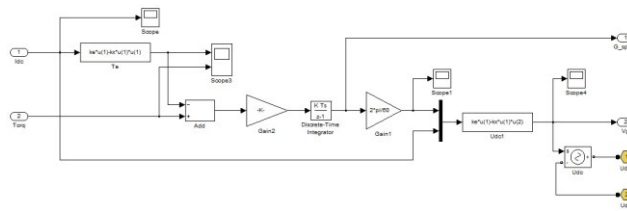


Figure5.Model of the Generator-rectifier

Dynamic Model of the Motor. The driving motor system is regarded as a controllable current source [5]. The model is shown in Fig.6.

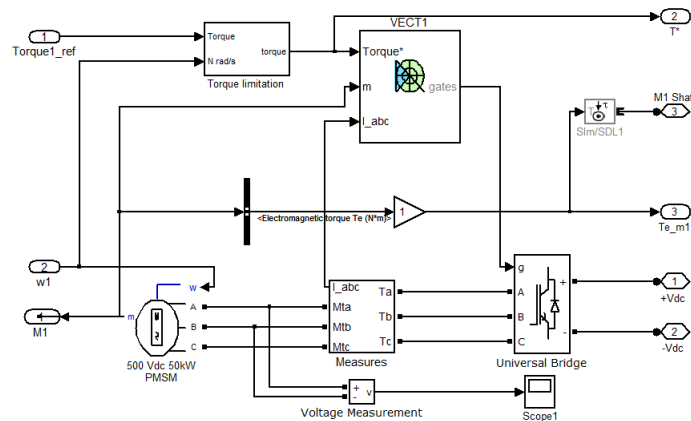


Figure6.Model of the Motor

The above models form the hybrid power system for electric bulldozer, as shown in Fig.7.

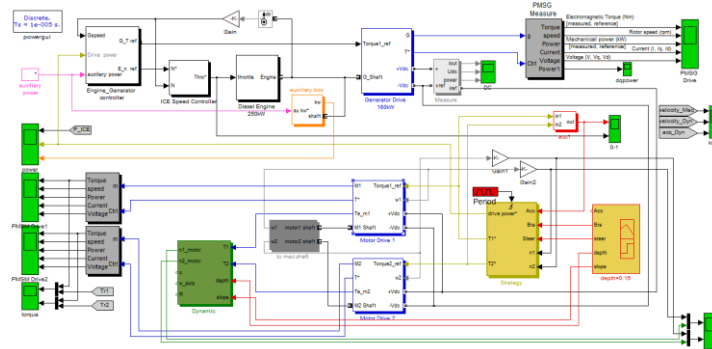


Figure7.Model of the hybrid power system for electric bulldozer

Simulation Experiments

According to the typical working characteristic, the above hybrid power system is simulated. The fuel consumption rate (be) comparison between traditional and hybrid power system is shown in fig.8.

As shown in fig.8, compared with traditional internal-combustion driving track-type bulldozers, hybrid power system bulldozer can yield greater fuel efficiency by nearly 11%.

Engine's torque and speed are shown in fig.9. In this simulation, engine operating point is controlled at the fuel consumption lowest point. The engine's torque and speed are controlled at 900Nm and 1200rpm.

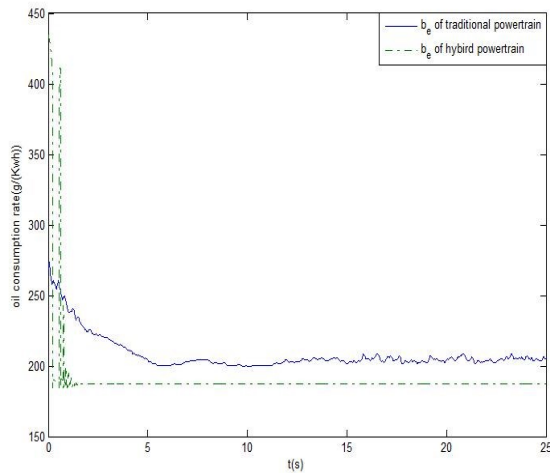


Figure8

Figure8.Fuel consumption comparison between traditional and hybrid power system

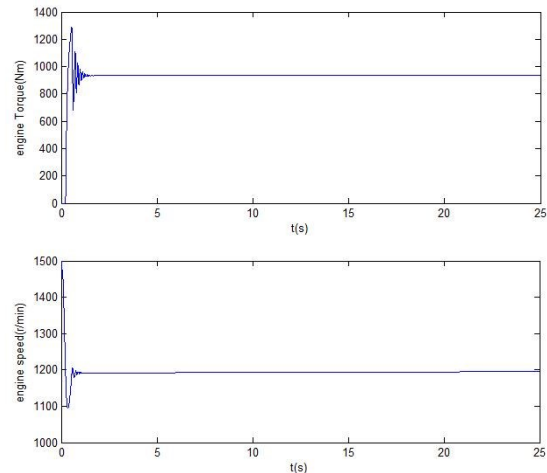


Figure9

Figure9.Engine's torque and speed

Conclusion

In this paper, the motor's mathematical model for dual-motor-drive electric bulldozer is established. The simulation results show:

- (1) Simulation results show that matching of the parameters of the hybrid power system is reasonable.
- (2) Simulation results demonstrate the validity of the hybrid power system model.
- (3) Compared with traditional internal-combustion driving track-type bulldozers, hybrid power system bulldozer can yield greater fuel efficiency by nearly 10%.
- (4) Reasonable matching of the parameters of the hybrid power system is necessary to get a higher performance.

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